Linear Algebra II HW 10

1. Show that if B is a real symmetric invertible matrix, then B^2 is positive definite.

Solution

Let λ be an eigenvalue of B. Then λ^2 is an eigenvalue of B^2. Since B is invertible, $\lambda \neq 0$. Hence $\lambda^2 > 0$ by Theorem 6.4.7.

2. Let $\begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$.

(a) Show that A is positive definite and that $\boldsymbol{x}^T A\boldsymbol{x} = \boldsymbol{x}^T B\boldsymbol{x}$ for all $\boldsymbol{x} \in \boldsymbol{R}^2$.

(b) Show that B is positive definite, but B^2 is not positive definite.

Solution

(a) A is positive definite since A is symmetric and its eigenvalues $\lambda_1 = \frac{1}{2}$, $\lambda_2 = \frac{3}{2}$ are both positive. If $\mathbf{x} \in \mathbb{R}^2$, then

$$\mathbf{x}^T A\mathbf{x} = x_1^2 - x_1x_2 + x_2^2 = \mathbf{x}^T B\mathbf{x}$$

(b) If $\mathbf{x} \neq \mathbf{0}$, then

$$\mathbf{x}^T B\mathbf{x} = \mathbf{x}^T A\mathbf{x} > 0$$

since A is positive definite. Therefore B is also positive definite. However,

$$B^2 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$$

is not positive definite. Indeed if $\mathbf{x} = (1, 1)^T$, then

$$\mathbf{x}^T B^2\mathbf{x} = 0$$

3. For each of the following functions, find the stationary point and determine the stationary point corresponds to a local minimum, local maximum, or saddle point:

(a) $f(x,y)=\frac{y}{x^2}+\frac{x}{y^2}+xy$

(b) $f(x,y,z)=x^3+xyz+y^2-3x$

Solution

(a) Stationary point = (1,1)

The Hessian of f is at $(1,1)$ is

$$\begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}$$

Its eigenvalues are $\lambda_1 = 9$, $\lambda_2 = 3$. Since both are positive, the matrix is positive definite and hence $(1,1)$ is a local minimum.

(b) Stationary point = (1,0,0) and (−1,0,0)

The Hessian of f at $(1,\ 0,\ 0)$ is

$$\begin{pmatrix} 6 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Its eigenvalues are $\lambda_1 = 6$, $\lambda_2 = 1+\sqrt{2}$, $\lambda_3 = 1-\sqrt{2}$. Since they differ in sign, $(1,\ 0,\ 0)$ is a saddle point.

The Hessian of f at $(-1,\ 0,\ 0)$ is

$$\begin{pmatrix} -6 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

Its eigenvalues are $\lambda_1 = -6$, $\lambda_2 = 1+\sqrt{2}$, $\lambda_3 = 1-\sqrt{2}$. Since they differ in sign, $(-1,\ 0,\ 0)$ is a saddle point.

4. For each of the following matrices, compute the determinants of all the leading principal submatrices and use them to determine whether the matrix is positive definite:

(a) $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$

(b) $\begin{pmatrix} 3 & 4 \\ 4 & 2 \end{pmatrix}$

(c) $\begin{pmatrix} 6 & 4 & -2 \\ 4 & 5 & 3 \\ -2 & 3 & 6 \end{pmatrix}$

(d) $\begin{pmatrix} 4 & 2 & 1 \\ 2 & 3 & -2 \\ 1 & -2 & 5 \end{pmatrix}$

Solution

(a) $\det(A_1) = 2$, $\det(A_2) = 3$, positive definite;

(b) $\det(A_1) = 3$, $\det(A_2) = -10$, not positive definite;

(c) $\det(A_1) = 6$, $\det(A_2) = 14$, $\det(A_3) = -38$, not positive definite;

(d) $\det(A_1) = 4$, $\det(A_2) = 8$, $\det(A_3) = 13$, positive definite

5. Let A be a symmetric $n \times n$ matrix and $\boldsymbol{x}$ be an $n \times 1$ column vector. Suppose $\boldsymbol{x}, A\boldsymbol{x}, \cdots, A^{n-1}\boldsymbol{x}$ are linearly independent and $A^n\boldsymbol{x} = -a_{n-1}A^{n-1}\boldsymbol{x} - \cdots - a_1 A\boldsymbol{x} - a_0\boldsymbol{x}$.

(a) Show that A is similar to $B = \begin{pmatrix} -a_{n-1} & \cdots & -a_1 & -a_0 \\ 1 & 0 & \cdots & 0 \\ & \ddots & \ddots & \vdots \\ \mathbf{0} & & 1 & 0 \end{pmatrix}_{n\times n}$.

(b) Suppose $1 + a_{n-1} + \cdots + a_1 + a_0 = 0$. Show that 1 is an eigenvalue. Find an eigenvector of A corresponding to eigenvalue 1.

Solution

(a) $$A\begin{pmatrix} | & & | & | \\ A^{n-1}\boldsymbol{x} & \cdots & A\boldsymbol{x} & \boldsymbol{x} \\ | & & | & | \end{pmatrix} = \begin{pmatrix} | & & | & | \\ A^{n}\boldsymbol{x} & \cdots & A^{2}\boldsymbol{x} & A\boldsymbol{x} \\ | & & | & | \end{pmatrix}$$

$$= \begin{pmatrix} | & & | & | \\ A^{n-1}\boldsymbol{x} & \cdots & A\boldsymbol{x} & \boldsymbol{x} \\ | & & | & | \end{pmatrix} \begin{pmatrix} -a_{n-1} & 1 & & 0 \\ \vdots & 0 & \ddots & \\ -a_1 & \vdots & \ddots & 1 \\ -a_0 & 0 & \cdots & 0 \end{pmatrix}_{n\times n}$$

$$\begin{pmatrix} | & & | & | \\ A^{n-1}\boldsymbol{x} & \cdots & A\boldsymbol{x} & \boldsymbol{x} \\ | & & | & | \end{pmatrix}^{T} A^{T} = \begin{pmatrix} -a_{n-1} & \cdots & -a_1 & -a_0 \\ 1 & 0 & \cdots & 0 \\ & \ddots & \ddots & \vdots \\ 0 & & 1 & 0 \end{pmatrix}_{n\times n} \begin{pmatrix} | & & | & | \\ A^{n-1}\boldsymbol{x} & \cdots & A\boldsymbol{x} & \boldsymbol{x} \\ | & & | & | \end{pmatrix}^{T}$$

Let $M = \begin{pmatrix} | & & | & | \\ A^{n-1}\boldsymbol{x} & \cdots & A\boldsymbol{x} & \boldsymbol{x} \\ | & & | & | \end{pmatrix}^{T}$. Since $\boldsymbol{x}, A\boldsymbol{x}, \cdots, A^{n-1}\boldsymbol{x}$ are linearly independent, M is invertible. We have $A = A^{T} = M^{-1}BM$

(b) Since A is similar to B, by Theorem 6.1.14, the characteristic polynomial of A and B are

$$p_A(\lambda) = p_B(\lambda) = (-1)^n(\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0).$$

Clearly, $p_A(1) = p_B(1) = (-1)^n(1 + a_{n-1} + \cdots + a_1 + a_0) = 0$. $\lambda = 1$ is an eigenvalue of A and B.

By Theorem 6.1.14, $(1,1,\cdots,1)^T$ is an eigenvector of B corresponding to eigenvalue 1. By Theorem 6.1.12, $M^{-1}(1,1,\cdots,1)^T$ is an eigenvector of A corresponding to eigenvalue 1.